

Blue Brain: World's First Virtual Brain



Ranjith S. is B.Tech in applied electronics and instrumentation from Vedaayasa Institute of Technology

Blue Brain is the world's first virtual brain. It can think, take decisions and store anything in memory, just as the human brain does.

Blue Brain project is the first comprehensive attempt at reverse-engineering the brain of a mammalian, so that through detailed simulations, functions of the brain can be understood. One of the main advantages of the project is brain uploading—even after a person dies, his knowledge will not be lost. Blue Brain can function exactly the same way as a human brain.

This project is also used to understand various brain diseases. The development of this project allows scientists to

understand the human brain in more detail. Blue Brain project was founded by Prof. Henry Markram from Brain and Mind Institute at EPFL (École Polytechnique Fédérale de Lausanne) in collaboration with IBM. The simulation does not consist simply of an artificial neural network but involves a biologically-realistic model of neurons. It is expected to shed light on the nature of consciousness.

World's first virtual brain

A virtual brain is a machine that mimics the operations of a mammalian brain and produces output. This operation requires a powerful supercomputer. Blue Gene is a supercomputer developed by IBM to handle such situations, hence the name Blue Brain.

It could be argued that artificial intelligence, robotics and even the most-advanced computational neuroscience approaches that have been used to model brain functions are merely if-then-like conditions in various forms. This project will use adaptation and learning algorithms as these systems are more powerful.

Reconstructing Blue (neocortical) column

To reconstruct Blue column, the three-dimensional morphology of the neuron is generated and the errors corrected. The corrected neurons are placed in a database to produce numerous clones. For the action-potential passage, ion channels are inserted in each neuron.

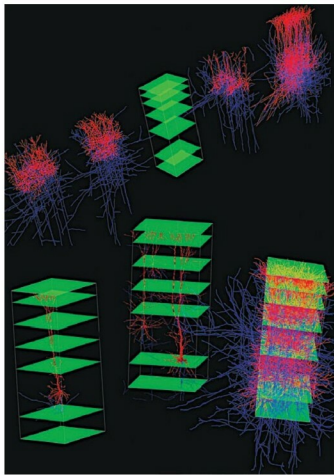


Fig. 1: Reconstructing the neocortical column